LEVEL UP - JEE ADVANCED 2022

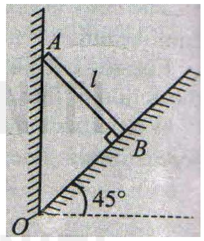
ROTATION

EDUNITI

SUBJECTIVE QUESTIONS

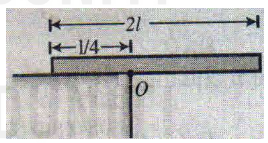
1. The door of an almirah is 3 m high, 1 m wide and weighs 30 kg. The door is supported by two hinges situated at a distance of 0.5 m from the ends. If the magnitude of each force exerted by the hinges on the door is equal, find this magnitude.

2. At the bottom edge of a smooth wall, an inclined plane is kept at an angle of 45^o. A uniform ladder of length and mass M rests on the incline plane against the wall such that it is perpendicular to the incline.

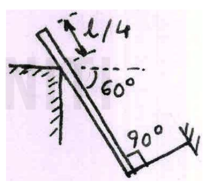


(a) If the plane is also smooth, which way will the ladder slide?
(b) What is the minimum coefficient of friction necessary so that the ladder does not slip on the incline.

3. One fourth length of a uniform rod of length $2l$ and mass m is placed on a horizontal table and the rod is held horizontal. The rod is released from rest. Find the normal reaction on the rod as soon as the rod is released.

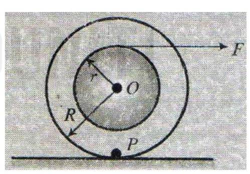


4. A uniform rod of mass m and length l is in equilibrium under the action of constraint forces, gravity and tension in the string. Find the



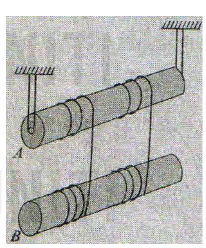
(a) frictional force acting onthe rod.
(b) tension in the string.
(c) normal reaction on the rod. Now the string is cut. Find the
(d) angular acceleration of the rod just after the string is cut
(e) normal reaction the rod just after the string is cut.

5. A cotton reel of mass m and moment of inertia I is kept at rest on a smooth horizontal surface. The reel has inner and outer radius r and R respectively. A horizontal force F starts acting as shown in fig. Find the

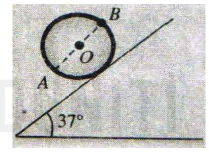


(a) acceleration of the centre of mass of reel.
(b) angular acceleration of the reel
(c) net acceleration of point of contact.

6. The arrangement shown in fig, consists of two identical, uniform, solid cylinders, each of mass m, on which two light threads are wound symmetrically. Find the tensions of each thread in the process of motion. The friction in the axle of the upper cylinder is assumed to be absent.

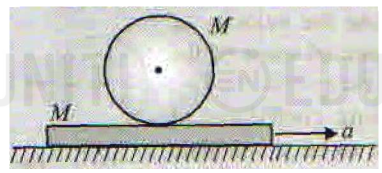


7. A uniform solid sphere of mass 1 kg and radius 10 cm is kept stationary on a rough inclined plane by fixing a highly dense particle at B. Inclination of plane is 37^o with horizontal and AB is the diameter of the sphere which is parallel to the plane, as shown in figure. Calculate

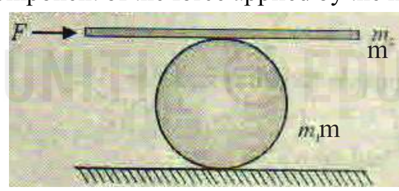


(a) mass of the particle at B
(b) minimum required coefficient of friction between sphere and plane to keep sphere in equilibrium.

8. A uniform cylinder (mass M) of radius R is kept on an accelerating platform (mass M) as shown in fig. If the cylinder rolls without sliping on the platform, determine the magnitude of acceleration of the centre of mass of the cylinder. Assuming the coefficient of friction $\mu = 0.40$, determine the maximum acceleration the platform may have without slipping between the cylinder and the platform.

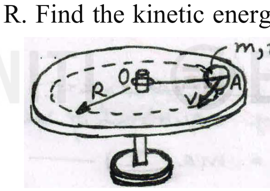


9. A man pushes a cylinder of mass m_1 with the help of a plank of mass m_2 as shown in fig. There is no slipping at any contact. The horizontal component of the force applied by the man is F. Find

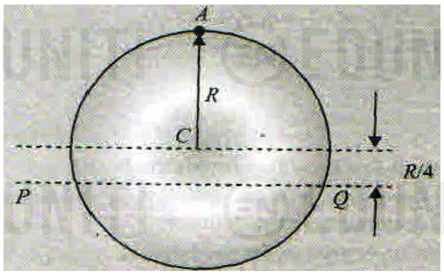


(a) the acceleration of the plank and the centre of mass of the cylinder.
(b) the magnitudes and directions of frictional forces at contact points.

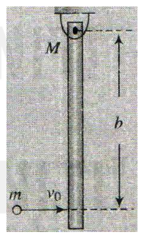
10. A uniform sphere of mass m and radius r rolls without sliding over a horizontal plane, rotating about a horizontal axis OA. In the process, the center of the sphere moves with a velocity v along a circle of radius R. Find the kinetic energy of the sphere.



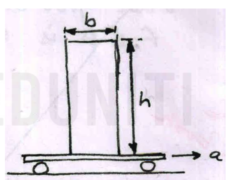
11. A uniform circular disc has radius R and mass m. A particle, also of mass m, is fixed at point A on the edge of the disc as shown in fig. The disc can rotate freely about a fixed horizontal chord PQ that is at a distance R/4 from the centre C of the disc. The line AC is perpendicular to PQ. Initially, the disc is held vertical with point A at its highest position. It is then allowed to fall so that it starts rotating about PQ. Find the linear speed of the particle as it reaches its lowest position.



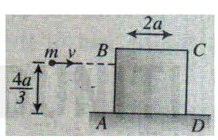
12. A ball of mass m collides elastically with a smooth hanging rod of mass M and length l.
(a) If M = 3m, find the value b for which no horizontal reaction occurs at the pivot.
(b) For what value of m/M, the ball falls dead just after the collision assuming e = 1/2



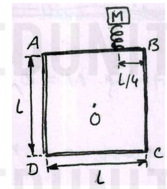
13. A block of mass m, height h and width b rests on a flat car which moves horizontally with constant acceleration 'a' as shown in figure.
Determine.
(a) the value of the acceleration at which slipping of the block on the car starts, if the coefficient of friction is μ
(b) the value of the acceleration at which block topples about A, assuming sufficient friction to prevent slipping and
(c) the shortest distance in which it can be stopped from a speed of 20 m s^{-1} with constant deceleration so that the block is not disturbed.
The following data are given
b = 0.2m, h = 0.5 m, μ = 0.5 and g = 10 m s^{-2}



14. A solid cube of wood of side 2a and mass M is resting on a horizontal surface. The cube is constrained to rotate about an axis passing through A and perpendicular to face ABCD. A bullet of mass m and speed v is shot at a height of 4a/3 as shown in the figure. The bullet becomes embedded in the cube. Find the minimum value of v required to topple the cube. Assume m << M.



15. A square frame is formed by four rods, each of length l = 60 cm. Mass of two rods AB and BC is m_1 = 25/18 kg each while that of rods AD and CD is m_2 = 2kg each. The frame is free to rotate about a fixed horizontal axis passing through its geometric centre O as shown in figure. A spring is placed on the rod AB at a distance l/4 = 15 cm from B. The spring is held vertical and a block is placed on upper end of the spring so that rod AB is horizontal.

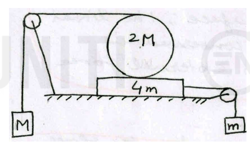


(a) Calculate mass M of the block
(b) If the spring is initially compressed by connecting a thread between its ends and energy stored in it is 76.5J, calculate velocity with which block bounces up when the thread is burnt.

PARAGRAPH TYPE QUESTIONS

Paragraph for Question Nos. 16 to 18

Figure shows a uniform smooth solid cylinder A of radius 4 m rolling without slipping on the 8 kg plank which, in turn is supported by a fixed smooth surface. Block B is known to accelerate down with 6 m/s^2 and block C moves down with acceleration 2 m/s^2



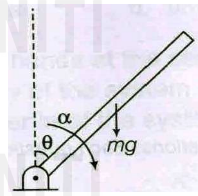
16. What is the angular acceleration of the cylinder
(a) 4 / 5 rad s^{-2} (b) 6 / 5 rad s^{-2}
(c) 2 rad s^{-2} (d) 1 rad s^{-2}

17. What is the ratio of the mass of the cylinder to the mass of block B?
(a) 1 (b) 2
(c) 3 (d) 4

18. If unwrapped length of the thread between the cylinder and block B is 20 m at the beginning, when the system was released from rest, what would it be 2 s later?
(a) 28 m (b) 30 m
(c) 22 m (d) 32.5 m

Paragraph for Question Nos. 19 to 21

A uniform rod of length L and mass m is pivoted freely at one end as shown in the figure. Answer the following questions.



19. The angular acceleration of the rod when it is at an angle θ to the vertical is

(a) $\frac{2g\sin\theta}{3L}$ (b) $\frac{g\sin\theta}{3L}$

(c) $\frac{4g\sin\theta}{3L}$ (d) $\frac{3g\sin\theta}{2L}$

20. Assuming the rod to start from the vertical position, the angular velocity as a function of θ is

(a) $\omega=\sqrt{\frac{3g}{L}(1-\sin\theta)}$ (b) $\omega=\sqrt{\frac{2g}{L}(1-\sin\theta)}$

(c) $\omega=\sqrt{\frac{3g}{L}(1-\cos\theta)}$ (d) $\omega=\sqrt{\frac{2g}{L}(1-\cos\theta)}$

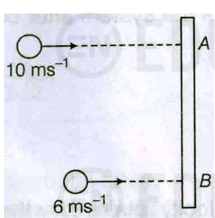
21. The tangential linear acceleration of the free end when the rod is horizontal, is

(a) 2g/3 (b) 3g/2

(c) 4g/3 (d) g/2

Paragraph for Question Nos. 22 to 24

A thin uniform bar lies on a frictionless horizontal surface and is free to move in any way on the surface. Its mass is 0.16 kg and length is $\sqrt{3}m$. Two particles, each of mass 0.08 kg are moving on the same surface and towards the bar in a direction perpendicular to the bar, one with a velocity 10 ms^{-1} and the other with 6 ms^{-1} as shown in the figure. The first particle strikes the bar at point A and the other at point B. Each of A and B is at a distance of 0.5 m from the centre of the bar. The particles strike the bar at the same instant of time and stick to the bar after collision.



22. The velocity of centre of mass of the system just after impact (in ms^{-1}) is

(a) 1 (b) 2

(c) 3 (d) 4

23. The angular velocity of the system just after impact (in rad/s) is

(a) 8 (b) 4

(c) 2 (d) 1

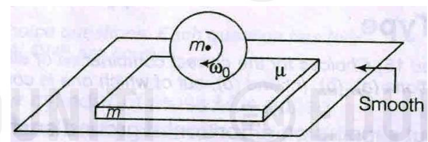
24. The loss in kinetic energy of the system in the above collision process is

(a) 2.72 J (b) 1.72 J

(c) 3.36 J (d) 4.36 J

Paragraph for Question Nos. 25 to 27

A long horizontal plank of mass m is lying on a smooth horizontal surface. A sphere of same mass m and radius R is spinned about its own axis with angular velocity ω_0 and gently placed on the plank. The coefficient of friction between the plank and the sphere is μ. After some time, pure rolling of the sphere on the plank starts.



25. The time after which pure rolling begins is

(a) $\frac{R\omega_0}{9\mu g}$ (b) $\frac{2R\omega_0}{9\mu g}$

(c) $\frac{R\omega_0}{3\mu g}$ (d) $\frac{2R\omega_0}{\mu g}$

26. The velocity of the sphere after pure rolling begins is

(a) $\frac{2}{9}R\omega_0$ (b) $\frac{R\omega_0}{9}$

(c) $\frac{R\omega_0}{3}$ (d) $\frac{2}{3}R\omega_0$

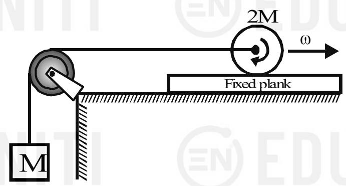
27. The displacement of the plank till the sphere starts pure rolling is

(a) $\frac{R^2\omega_0^2}{81\mu g}$ (b) $\frac{2}{27}\left[\frac{R^2\omega_0^2}{\mu g}\right]$

(c) $\frac{4}{81}\left[\frac{R^2\omega_0^2}{\mu g}\right]$ (d) $\frac{2}{81}\left[\frac{R^2\omega_0^2}{\mu g}\right]$

Paragraph for Questions Nos. 28 to 30

A disc of mass 2 M and radius R is placed on a fixed plank (rough) of length L. The coefficient of friction between the plank and disc is $\mu = 0.5$. A light string is connected to centre of disc and passes over a smooth light pulley and connected to a block of mass M as shown in the figure. Now the disc is given an angular velocity ω_0 in clockwise direction and is gently placed on the plank at its right end. Consider this instant as t = 0. Based on above information, answer the following questions



28. Mark the correct statement w.r.t. motion of block and disc.
    (a) The block remains at rest for some time, $t > 0$.
    (b) The block starts accelerating just after placing of disc on plank.
    (c) The disc is performing pure rotational motion for some time $t > 0$
    (d) Both (a) and (c) are correct.

29. Time t_0 upto which the block remains stationary is

    (a) $\frac{\omega_0 R}{g}$ (b) $\frac{4\omega_0 R}{g}$

    (c) Zero (d) Question is irrelevant

30. Time time t_1 at which the disc will cross the other end of the plank is

    (a) $\sqrt{\frac{8L}{g}}$ (b) $\frac{\omega_0 R}{g} + \sqrt{\frac{8L}{g}}$

    (c) $\sqrt{\frac{8L}{g}} + \frac{4\omega_0 R}{g}$ (d) $\sqrt{\frac{\omega_0 R}{g}} + \sqrt{\frac{4L}{g}}$